

Module -5

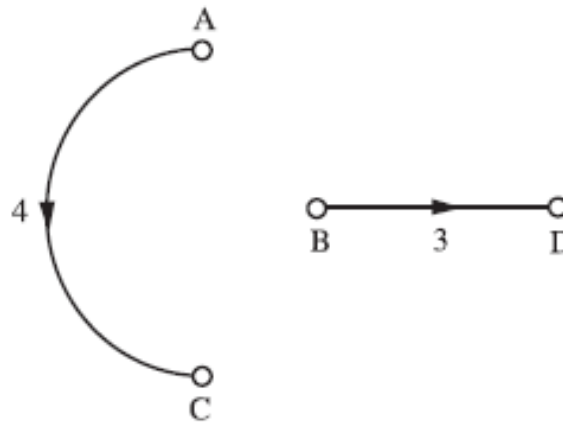
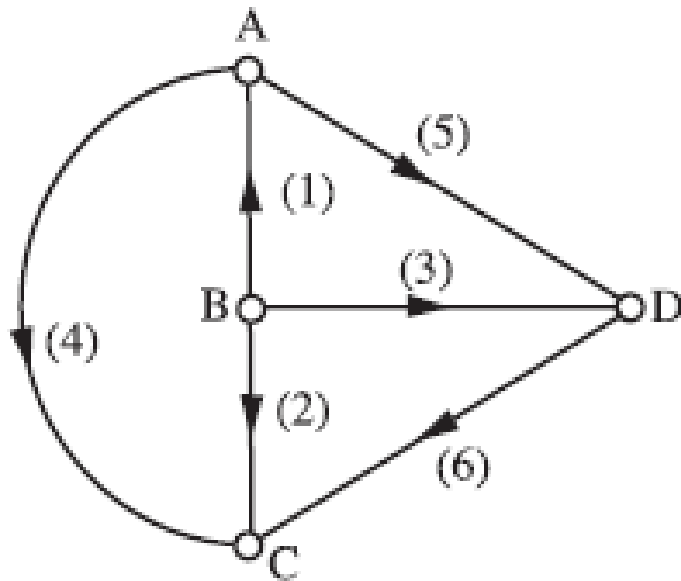
Fundamental Cut-set Matrix

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Fundamental Cut-Set

- Tree branches connect all the nodes in the network graph. It is possible to trace the path from one node to any other node by travelling along the tree branch only.
- Therefore, potential difference between any two nodes called node – pair voltages can be expressed in terms of tree branch voltages.
- The cut set is a minimal set of branches of the graph, removal of which cuts the graph into two parts. It separates the nodes of the graph into two groups.

- The cut set consists of only one tree branch and remainders are links.
- Each branch of the cut-set has one of its terminal incident at a node in one group and its other end at a node in the other group.
- The orientation of the cut-set is same as orientation of tree branch.
- The number of cut-sets is equal to the number of tree branches ($N-1$).



Two separate graphs created by
the cut set {1, 2, 5, 6}

Cut-Set Matrix

- This matrix has current values and dimension is $(N-1) \times B$.

1. Draw the graph of given network or circuit.
2. Then draw its tree. The branches of the tree will be twig.
3. Then draw the remaining branches of the graph by dotted line. These branches will be links.
4. Assume the tree branch voltages as $(e_1, e_2, \dots, e_n)$ independent variables.
5. Each branch or twig of tree will form an independent cut-set.
6. Write the matrix with rows as cut-set and column as branches.

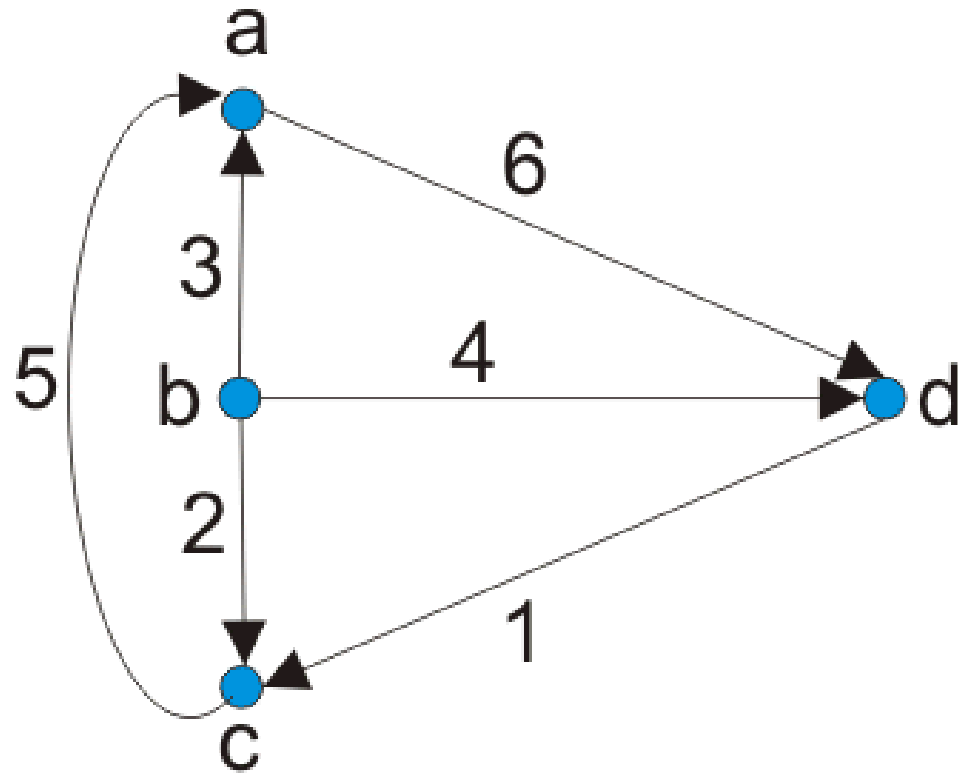
$Q_{ij} = 1$; if branch J is in cut-set with orientation same as that of tree branch.

$Q_{ij} = -1$; if branch J is in cut-set with orientation opposite to that of branch of tree.

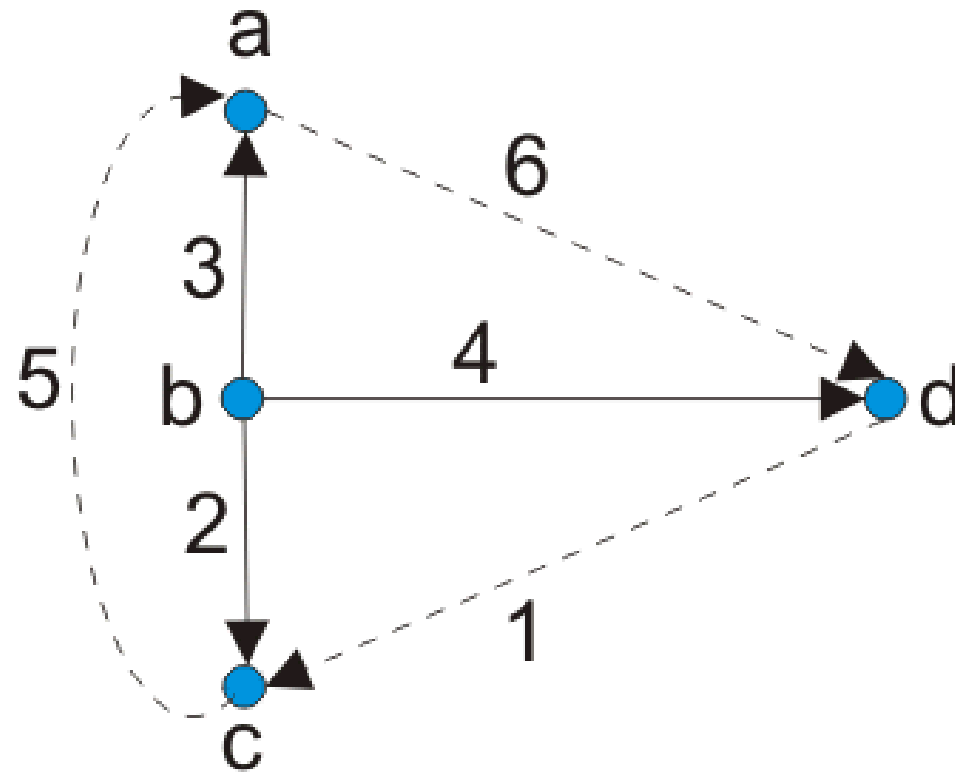
$Q_{ij} = 0$; if branch J is not in cut-set.

Numerical - 1

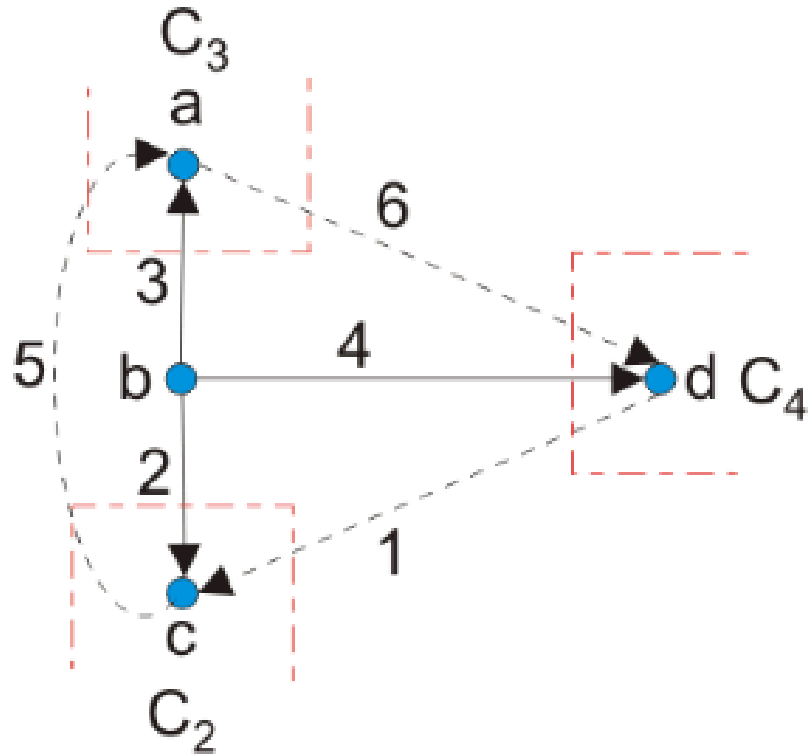
Draw the cut-set matrix for the following graph.



Step 1: Draw the tree for the following graph.



- Step 2: Now identify the cut-set. Cut-set will be that node which will contain only one twig and any number of links.



Branches						
	v1	v2	v3	v4	v5	v6
	1	2	3	4	5	6
e_2	1	1	0	0	-1	0
e_3	0	0	1	0	1	-1
e_4	-1	0	0	1	0	1

The columns of the cut-set schedule give branch voltage equations:

$$\begin{aligned} V_1 &= e_2 - e_4 \\ V_2 &= e_2 \\ V_3 &= e_3 \\ V_4 &= e_4 \\ V_5 &= e_3 - e_2 \\ V_6 &= e_4 - e_3 \end{aligned} \quad \left. \vphantom{\begin{aligned} V_1 &= e_2 - e_4 \\ V_2 &= e_2 \\ V_3 &= e_3 \\ V_4 &= e_4 \\ V_5 &= e_3 - e_2 \\ V_6 &= e_4 - e_3 \end{aligned}} \right\} \text{Eqn. (1)}$$

The rows of the cut-set schedule KCL equations:

$$\begin{aligned} j_1 + j_2 - j_5 &= 0 \\ j_3 + j_5 - j_6 &= 0 \\ -j_1 + j_4 + j_6 &= 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} j_1 + j_2 - j_5 &= 0 \\ j_3 + j_5 - j_6 &= 0 \\ -j_1 + j_4 + j_6 &= 0 \end{aligned}} \right\} \text{Eqn. (2)}$$

- The cut-set schedule can be written in matrix form is known as cut-set matrix (Q).

$$Q = \begin{pmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

- From set of eqn. (1), $[V_B] = [Q]^T [E_N]$

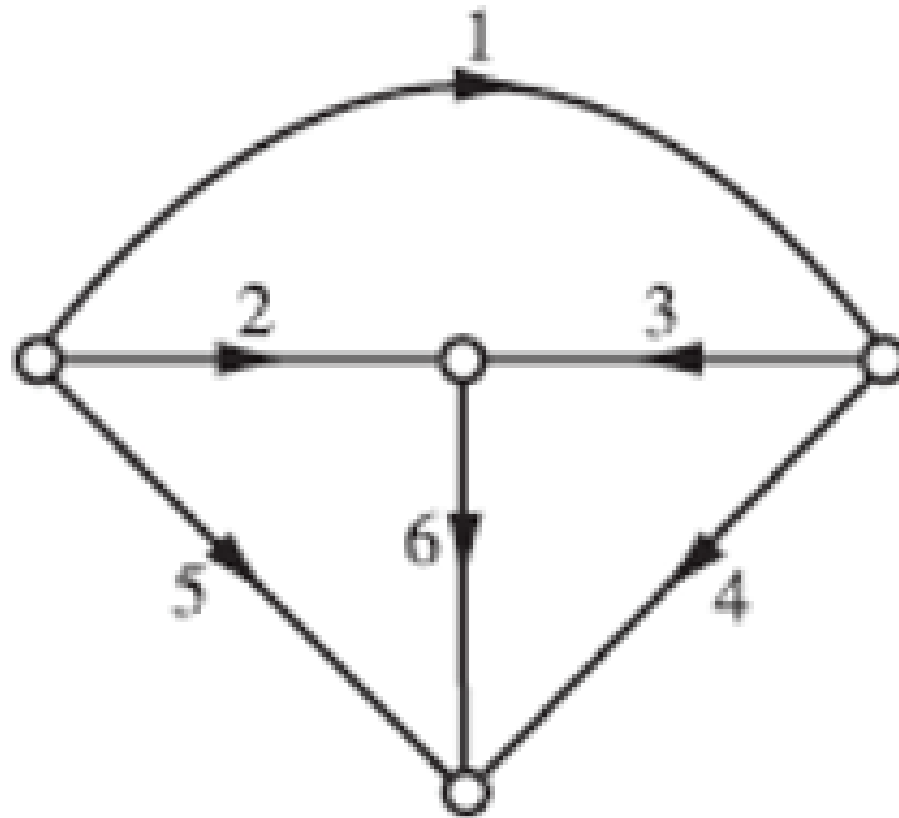
$$\begin{pmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \\ V_6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} e_2 \\ e_3 \\ e_4 \end{pmatrix}$$

- From set of eqn. (2), $[Q] [I_B] = 0$

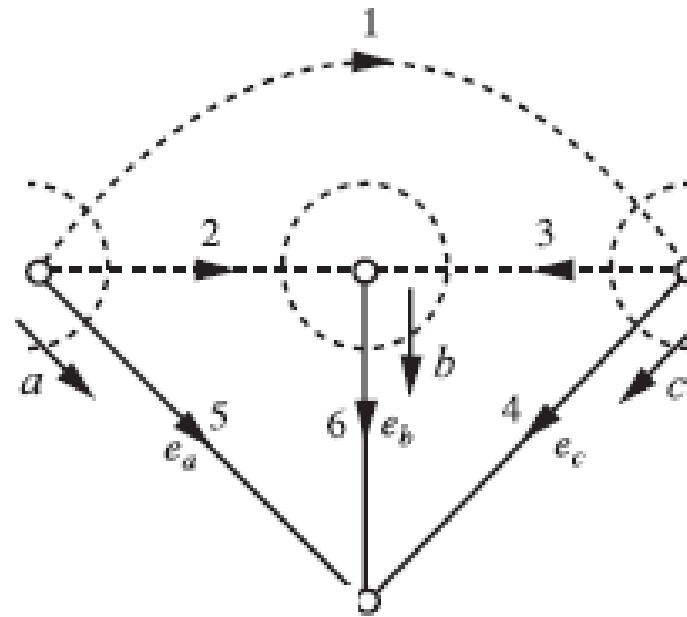
$$\begin{pmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} J_1 \\ J_2 \\ J_3 \\ J_4 \\ J_5 \\ J_6 \end{pmatrix} = 0$$

Numerical - 2

- Draw the cut-set matrix for the following graph.



Choosing 4, 5, 6 as tree branches,



$$\mathbf{Q} = \begin{matrix} e_a \\ e_b \\ e_c \end{matrix} \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \\ -1 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \quad \begin{aligned} J_1 + J_2 + J_5 &= 0 \\ -J_2 - J_3 + J_6 &= 0 \\ -J_1 + J_3 + J_4 &= 0 \end{aligned}$$

$$v_1 = e_a - e_c; \quad v_2 = e_a - e_b; \quad v_3 = e_b - e_c; \quad v_4 = e_c; \quad v_5 = e_a; \quad v_6 = e_b$$